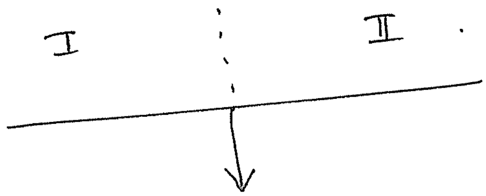
Solution Set 8

1

$$V(x) = -V_0 \lambda\, \delta(x)$$



Particle energy $E = \dfrac{\hbar^2 k^2}{2m}$. The wave function must satisfy the equation

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - V_0 \lambda\, \delta(x)\, \psi = \frac{\hbar^2 k^2}{2m}\psi .$$

Integrating on both sides from $-\varepsilon$ to ε (ε = small) we get

$$\lim_{\varepsilon \to 0} \left.\frac{d\psi}{dx}\right|_{+\varepsilon} - \left.\frac{d\psi}{dx}\right|_{-\varepsilon} = -\frac{2mV_0}{\hbar^2}\lambda\, \psi(0)$$

$\therefore$ If $\psi_I(x)$ is the wave function in region I and $\psi_{II}(x)$ is the wave function in region II.

$$\left.\frac{d\psi_{II}}{dx}\right|_{x=0} - \left.\frac{d\psi_I}{dx}\right|_{x=0} = -\frac{2mV_0}{\hbar^2}\lambda\, \psi_I(0)$$

and $\psi_I(0) = \psi_{II}(0)$ (Continuity)

B.C of the wavefunction.

Now

$$\psi_I(x) = A e^{ikx} + B e^{-ikx}$$

$$\psi_{II}(x) = C e^{ikx}$$

$\therefore$

$$A + B = C$$

$$Cik - (Aik - Bik) = -\frac{2mV_0}{\hbar^2}\lambda\, C$$

Let us define $\frac{2\mu V_0 \lambda}{\hbar^2} = k_0$, then we have

$$C(ik + k_0) = ik(A - B)$$

$$C = A + B$$

$$\therefore \quad A = C\left(1 + \frac{k_0}{2ik}\right)$$

$$B = -\frac{C k_0}{2ik}.$$

$$R = \frac{|B|^2}{|A|^2} = \frac{k_0^2/4k^2}{(1 + k_0^2/4k^2)}.$$

$$T = \frac{|C|^2}{|A|^2} = \frac{1}{(1 + k_0^2/4k^2)}.$$

(2) The probability current $\vec{J}_{sc}$ is defined as

$$\vec{J}_{sc} = \frac{\hbar}{2\mu i}\left(\psi_{sc}^* \vec{\nabla} \psi_{sc} - \nabla \psi_{sc}^* \psi_{sc}\right).$$

where $\psi_{sc} = \frac{e^{\pm ikr}}{r} f(\theta, \phi)$.

In spherical coordinates

$$\vec{\nabla} = \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r}\frac{\partial}{\partial \theta} + \hat{e}_\phi \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\right).$$

In the $r \to \infty$ limit the $\hat{e}_\theta$ and $\hat{e}_\phi$ are subleading contributions and so can be neglected.

The leading contribution comes in the $\hat{e}_r$ direction when the "$\frac{\partial}{\partial r}$" acts on the $e^{\pm ikr}$ part of the wave function. Thus.

$$\vec{J}_{sc} = \pm \frac{\hbar k}{\mu} \frac{|f|^2}{r^2} \hat{e}_r + O(\text{higher orders})$$

The + sign leads to the outgoing current.

– " " incoming " .

(3) The free particle energy eigenfunctions can be chose to be simultaneous eigenfunctions of $\vec{L}^2$ and L_z.

since the problem is spherically symmetric.

$$\psi_{k\ell m}(\vec{r}) = R_\ell(kr)\, Y_\ell^m(\theta,\phi)$$

where the function $R_\ell(kr)$ satisfies the equation

$$-\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} - \frac{\ell(\ell+1)}{r^2}\right] R_\ell(kr) = \frac{\hbar^2 k^2}{2\mu} R_\ell(kr)$$

(Shankar eq. 12.6.3)

where we assume $E = \frac{\hbar^2 k^2}{2\mu}$.

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} - \frac{\ell(\ell+1)}{r^2} + k^2\right] R_\ell(kr) = 0$$

(b) We see that $R_\ell(\rho)$ satisfies the eq.

$$\left[\frac{1}{\rho^2}\frac{\partial}{\partial\rho}\rho^2\frac{\partial}{\partial\rho} - \frac{\ell(\ell+1)}{\rho^2} + 1\right] R_\ell(\rho) = 0$$

The solutions are the spherical Bessel function $j_\ell(\rho)$ and spherical Neumann functions $\eta_\ell(\rho)$.

Note if we write $R_\ell(\rho) = \frac{U_\ell(\rho)}{\rho}$ then we get the eq.

$$\left[\frac{d^2}{d\rho^2} - \frac{\ell(\ell+1)}{\rho^2} + 1\right] U_\ell(\rho) = 0.$$

whose solutions are $\rho j_\ell(\rho)$ and $\rho \eta_\ell(\rho)$.

(c) The large distance behavior of $j_\ell(\rho)$ and $\eta_\ell(\rho)$ is given by.

$$\lim_{\rho\to\infty} j_\ell(\rho) = \frac{1}{\rho}\sin\left(\rho - \frac{\ell\pi}{2}\right)$$

$$\lim_{\rho\to\infty} \eta_\ell(\rho) = -\frac{1}{\rho}\cos\left(\rho - \frac{\ell\pi}{2}\right).$$

Thus these are <u>super positions</u> of spherically incoming and outgoing waves. To get an out going wave need to combine $j_\ell(\rho)$ and $\eta_\ell(\rho)$. But for the free particle $\eta_\ell(\rho)$ is not an allowed function since it blows up at the origin. So we cannot construct a purely outgoing spherical wave for the free particle! (Indeed that would imply there is a source at the center!)

19.3.1 In the class we showed that if

$$V(r) = \frac{g\, e^{-\mu_0 r}}{r}$$

then

$$\frac{d\sigma}{d\Omega} = \frac{4\mu^2 g^2}{\hbar^4 [\mu_0^2 + 4k^2 \sin^2 \theta/2]^2}$$

To obtain the total scatteng cross section we need

$$\sigma = \int \frac{d\sigma}{d\Omega}\, d\Omega = 2\pi \int_0^\pi d\theta \sin\theta \frac{4\mu^2 g^2}{\hbar^4[\mu_0^2 + 2k^2 - 2k^2\cos\theta]^2}$$

$$= 16\pi r_0^2 \left(\frac{g\mu r_0}{\hbar^2}\right)^2 \frac{1}{(1+4k^2 r_0^2)}.$$

where $r_0 = \frac{1}{\mu_0}$. Compare this with the

Geometric cross section $\simeq \pi r_0^2$

19.3.2 Given a spherically symmetric potential $V(r)$ we found in class (see (eq. 19.3.8) is Shankar)

$$f(\theta) = -\frac{2\mu}{\hbar^2 q}\int_0^\infty \sin qr'\, V(r')\, r'\, dr'$$

Given $V(r) = -V_0\,\theta(r_0 - r)$ we see that

$$f(\theta) = +\frac{2\mu V_0}{\hbar^2 q}\int_0^{r_0} r' \sin qr'\, dr'$$

$$= \frac{2\mu V_0}{\hbar^2 q}\left\{\frac{(\sin qr_0 - (qr_0)\cos qr_0)}{q^2}\right\}$$

$\therefore$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = 4r_0^2\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2\frac{(\sin qr_0 - (qr_0)\cos qr_0)^2}{(qr_0)^6}$$

where $q^2 = 2k^2(1-\cos\theta)$.

As $kr_0 \to 0$ $qr_0 \to 0$ we see that

$$\frac{(\sin qr_0 - (qr_0)\cos qr_0)^2}{(qr_0)^6} \simeq \frac{1}{9}\left[1 - O(qr_0)^2\right]$$

$$\therefore \quad \sigma \simeq \frac{16\pi r_0^2}{9}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2$$

19.3.3 For the Gaussian Potential $V(r) = V_0 e^{-r^2/r_0^2}$

$$f(\theta) = \frac{-2\mu V_0}{\hbar^2 q}\int_0^\infty r' \sin qr' \, e^{-r'^2/r_0^2} dr'$$

$$= \left(-\frac{2\mu V_0}{\hbar^2 q}\right)\left(-\frac{\partial}{\partial q}\right)\int_0^\infty \underbrace{\cos qr'}_{\frac{e^{iqr'}+e^{-iqr'}}{2}} e^{-r'^2/r_0^2} dr'$$

$$\left[\int_0^\infty e^{-iqr'} e^{-r'^2/r_0^2} dr' = \int_{-\infty}^0 e^{iqr'} e^{-r'^2/r_0^2} dr'\right]$$

$$= \left(\frac{-2\mu V_0}{\hbar^2 q}\right)\left(-\frac{\partial}{\partial q}\right)\left\{\frac{\sqrt{\pi r_0^2}}{2}\left(e^{-\frac{q^2 r_0^2}{4}}\right)\right\}$$

$$= \left(-\frac{2\mu V_0}{\hbar^2 \cancel{q}}\right)\sqrt{\pi r_0^2}\,\frac{1}{4}\,\cancel{q} r_0^2 e^{-q^2 r_0^2/4}$$

$$\frac{d\sigma}{d\Omega} = |f|^2 = \frac{\pi r_0^2}{4}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 e^{-q^2 r_0^2/2}$$

The total cross section.

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = (2\pi)\frac{\pi r_0^2}{4}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \int_0^\pi d\theta \sin\theta \, e^{-q^2 r_0^2/2}$$

Let $q^2 = 2k^2(1-\cos\theta)$

$$\frac{d(q^2)}{2k^2} = \sin\theta \, d\theta$$

Hence

$$\sigma = \frac{\pi^2 r_0^2}{2}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \underbrace{\int_0^{4k^2} \frac{d(q^2)}{2k^2}\, e^{-q^2 r_0^2/2}}_{\frac{2\left(1-e^{-2k^2 r_0^2}\right)}{2k^2 r_0^2}}$$

$$\therefore \quad \sigma = \frac{\pi^2}{2k^2}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \left(1 - e^{-2k^2 r_0^2}\right)$$